THE PROPERTY OF BEING PRO-ISOMORPHIC TO A DELIGNE SYSTEM SATISFIES 2-OUT-OF-3 ALONG DISTINGUISHED TRIANGLES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0G4Q (sources/stacks-duality/pinned-original-source.tex line 8236).

Allowed standard prerequisites (editorial).

- Essential constancy satisfies 2-out-of-3 along distinguished triangles. Derived Categories, Lemma labelled derived-lemma-essentially-constant-2-out-of-3, cited at proof lines 8245-8248. Gives essential constancy of the restriction of (L_n) to U.
- The colimit of Hom out of a Deligne system computes Hom on the open. Same chapter, Lemma labelled lemma-lift-map, cited at proof lines 8270-8272. Makes all but the middle vertical arrow of the diagram an isomorphism.
- Existence of a Deligne system with prescribed constant value, and the characterisation of pro-isomorphism by colimits of Hom. Same chapter, Lemma labelled lemma-extension-by-zero, cited at proof lines 8278-8280, and Categories, Remark labelled categories-remark-pro-category-copresheaves, cited at proof lines 8280-8281. Concludes that (L_n) is pro-isomorphic to a Deligne system.

Source reference. Original lemma 0G3A: sources/stacks-chow/tags.txt, source lines 20085–20085. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8251 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0G2H: sources/stacks-duality/pinned-original-source.tex, source lines 7920–7920. Same-chapter pointer into the author environment lemma at source line 7920; invoked at proof line 8270 of the counted proof.

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Source reference. Original lemma 0G4L: sources/stacks-duality/pinned-original-source.tex, source lines 8103–8103. Same-chapter pointer into the author environment lemma at source line 8103; invoked at proof line 8276 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0G2X: sources/stacks-chow/tags.txt, source lines 20072–20072. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8280 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $j : U \to X$ be an open immersion of Noetherian schemes. Let*

$$K_n \to L_n \to M_n \to K_n[1]$$

be an inverse system of distinguished triangles in $D^b_{Coh}(\mathcal{O}_X)$. If (K_n) and (M_n) are pro-isomorphic to Deligne systems, then so is (L_n).

Proof. Observe that the systems $(K_n|_U)$ and $(M_n|_U)$ are essentially constant as they are pro-isomorphic to constant systems. Denote K and M their values. By Derived Categories, Lemma 0G3A we see that the inverse system $L_n|_U$ is essentially constant as well. Denote L its value. Let $N \in D^b_{Coh}(\mathcal{O}_X)$. Consider the commutative diagram

$$\begin{array}{ccccccccc}
\ldots & \longrightarrow & \operatorname{colim} \operatorname{Hom}_X(M_n, N) & \longrightarrow & \operatorname{colim} \operatorname{Hom}_X(L_n, N) & \longrightarrow & \operatorname{colim} \operatorname{Hom}_X(K_n, N) & \longrightarrow & \ldots \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
\ldots & \longrightarrow & \operatorname{Hom}_U(M, N|_U) & \longrightarrow & \operatorname{Hom}_U(L, N|_U) & \longrightarrow & \operatorname{Hom}_U(K, N|_U) & \longrightarrow & \ldots
\end{array}$$

By Lemma 0G2H and the fact that isomorphic ind-systems have the same colimit, we see that the vertical arrows two to the right and two to the left of the middle one are isomorphisms. By the 5-lemma we conclude that the middle vertical arrow is an isomorphism. Now, if (L'_n) is a Deligne system whose restriction to U has constant value L (which exists by Lemma 0G4L), then we have $\operatorname{colim} \operatorname{Hom}_X(L'_n, N) = \operatorname{Hom}_U(L, N|_U)$ as well. Hence the pro-systems (L_n) and (L'_n) are pro-isomorphic by Categories, Remark 0G2X. □